Global vs. Per-Comparison Dispersion and Size Factor Estimation in Single-Cell CRISPR Screens

CRISPyx Development Team

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Abstract

We present a computational innovation for negative binomial generalized linear model (NB-GLM) analysis of single-cell CRISPR perturbation screens. Traditional differential expression pipelines (DESeq2, PyDESeq2) estimate dispersion parameters *per comparison*, which scales as $O(K)$ for K perturbations. We introduce **global dispersion estimation** that computes the dispersion trend once using all cells, reducing complexity to $O(1)$ per perturbation. This document details the biological motivation, mathematical formulation, and empirical validation showing perfect correlation ($\rho = 1.000$) between global and per-comparison approaches with **1.7–2.3× speedup** on real datasets.

Contents

1 Introduction: The CRISPR Screen Analysis Challenge

Single-cell CRISPR screens generate datasets with:

- $N \sim 10^4$–10^6 cells
- $G \sim 5{,}000$–$20{,}000$ genes
- $K \sim 10^2$–10^4 perturbations (guide RNA targets)

For each perturbation k, we fit an NB-GLM comparing treated cells to controls:

$$Y_{cg} \sim \mathrm{NB}(\mu_{cg}, \alpha_g)$$

where Y_{cg} is the UMI count for cell c and gene g, μ_{cg} is the expected count, and α_g is the gene-wise dispersion parameter.

1.1 The Dispersion Bottleneck

DESeq2/PyDESeq2 estimate dispersion using a three-step procedure:

1. **Gene-wise MLE**: Compute maximum likelihood estimate $\hat{\alpha}_g^{\mathrm{MLE}}$ for each gene
2. **Trend fitting**: Fit a parametric trend $\alpha(\bar{\mu}_g)$ relating dispersion to mean expression
3. **MAP shrinkage**: Shrink MLE toward trend using empirical Bayes (maximum a posteriori)

In multi-perturbation screens, this is repeated for *each of the K comparisons*, resulting in:

$$\text{Total dispersion cost} = O(K \cdot G \cdot N_{\mathrm{iter}})$$

For a screen with $K = 1{,}000$ perturbations, this becomes a major bottleneck.

2 Biological Motivation for Global Dispersion

2.1 Dispersion Reflects Technical Noise, Not Perturbation Effects

The dispersion parameter α_g in the NB model captures **overdispersion** relative to Poisson:

$$\mathrm{Var}(Y_{cg}) = \mu_{cg} + \alpha_g \mu_{cg}^2$$

This overdispersion arises from:

- **Technical noise**: PCR amplification bias, capture efficiency variation, sequencing depth differences
- **Biological heterogeneity**: Cell cycle state, transcriptional bursting

Critically, these sources of variation are **perturbation-independent**:

- A gene's PCR amplification efficiency doesn't change based on which guide RNA was delivered
- Transcriptional bursting kinetics are gene-intrinsic properties
- Library preparation artifacts affect all cells equally

Proposition 1 (Perturbation-Invariance of Dispersion)**.** *For gene g, let $\alpha_g^{(k)}$ denote the dispersion estimated from comparison k. Under the assumption that overdispersion reflects technical/intrinsic biological noise:*

$$\alpha_g^{(k)} \approx \alpha_g^{(k')} \quad \forall k, k' \in \{1, \ldots, K\}$$

This biological insight motivates pooling all cells for dispersion estimation.

2.2 Statistical Power from Increased Sample Size

Dispersion estimation quality improves with sample size. Using control cells alone (N_{ctrl}) vs. all cells (N_{total}):

$$\text{SE}(\hat{\alpha}_g) \propto \frac{1}{\sqrt{N}}$$

For a screen with $N_{\text{total}} = 100{,}000$ cells where controls comprise 5%:

$$\frac{\text{SE}_{\text{control-only}}}{\text{SE}_{\text{all-cells}}} = \sqrt{\frac{100{,}000}{5{,}000}} \approx 4.5$$

Global estimation provides 4.5× more precise dispersion estimates.

3 Mathematical Formulation

3.1 Negative Binomial GLM

For each gene g and cell c:

$$Y_{cg} \sim \text{NB}(\mu_{cg}, \alpha_g) \tag{1}$$

$$\log(\mu_{cg}) = \eta_{cg} = \mathbf{x}_c^\top \boldsymbol{\beta}_g + \log(s_c) \tag{2}$$

where:

- $\mathbf{x}_c \in \mathbb{R}^p$ is the design vector (intercept, perturbation indicator, covariates)
- $\boldsymbol{\beta}_g \in \mathbb{R}^p$ are gene-specific coefficients
- $s_c > 0$ is the size factor (normalization offset)
- $\alpha_g > 0$ is the dispersion parameter

The NB log-likelihood for gene g is:

$$\ell_g(\boldsymbol{\beta}_g, \alpha_g) = \sum_{c=1}^{N} \left[y_{cg} \log \frac{\alpha_g \mu_{cg}}{1 + \alpha_g \mu_{cg}} - \frac{1}{\alpha_g} \log(1 + \alpha_g \mu_{cg}) + \log \Gamma\left(y_{cg} + \frac{1}{\alpha_g}\right) - \log \Gamma\left(\frac{1}{\alpha_g}\right) \right] \tag{3}$$

3.2 Per-Comparison Dispersion (Standard DESeq2)

For comparison k (perturbation k vs. control), let $\mathcal{C}_k \subset \{1, \ldots, N\}$ denote the cell subset. The per-comparison approach:

1. **Subset selection**: Use only cells in comparison k: $\{c : c \in \text{control} \cup \text{perturbation}_k\}$
2. **Size factors**: Compute $s_c^{(k)}$ from subset (median-of-ratios on $|\mathcal{C}_k|$ cells)
3. **Dispersion MLE**: Maximize profile likelihood over subset:

$$\hat{\alpha}_g^{(k),\text{MLE}} = \arg\max_{\alpha > 0} \sum_{c \in \mathcal{C}_k} \ell_{cg}(\hat{\boldsymbol{\beta}}_g^{(k)}, \alpha) \tag{4}$$

4. **Trend fitting**: Fit $\alpha^{(k)}(\bar{\mu}) = a_0^{(k)} + a_1^{(k)}/\bar{\mu}$ using subset

5. **MAP shrinkage**:

$$\hat{\alpha}_g^{(k),\text{MAP}} = \arg\max_{\alpha>0} \left[\ell_g^{(k)}(\alpha) + \log p(\alpha \mid \sigma_{\text{prior}}^{(k)}) \right] \tag{5}$$

Complexity: Steps 1–5 repeated K times $\Rightarrow O(K \cdot G \cdot N_{\text{subset}})$

3.3 Global Dispersion (Our Innovation)

We compute dispersion *once* using all cells:

1. **Global size factors**: Compute s_c from all N cells
2. **Global dispersion MLE**: Use all cells for moments-based initialization:

$$\hat{\alpha}_g^{\text{global,init}} = \frac{\text{Var}(\tilde{Y}_{\cdot g}) - \bar{\mu}_g}{\bar{\mu}_g^2} \tag{6}$$

   where $\tilde{Y}_{cg} = Y_{cg}/s_c$ are normalized counts.

3. **Global trend**: Fit dispersion-mean trend on all cells:

$$\alpha^{\text{global}}(\bar{\mu}) = a_0^{\text{global}} + \frac{a_1^{\text{global}}}{\bar{\mu}} \tag{7}$$

4. **Global prior variance**: Estimate σ_{prior}^2 from global residuals:

$$\sigma_{\text{prior}}^2 = \text{Var}\left(\log \hat{\alpha}_g^{\text{MLE}} - \log \alpha^{\text{global}}(\bar{\mu}_g) \right) - \psi'(\text{df}/2) \tag{8}$$

   where ψ' is the trigamma function (bias correction for log-normal).

5. **Per-perturbation MAP** (optional refinement): For each comparison k:

$$\hat{\alpha}_g^{(k),\text{MAP}} = \arg\max_{\alpha>0} \left[\ell_g^{(k)}(\alpha) + \log p(\alpha \mid \alpha^{\text{global}}(\bar{\mu}_g), \sigma_{\text{prior}}^{\text{global}}) \right] \tag{9}$$

Complexity: Steps 1–4 computed once $O(G \cdot N)$; Step 5 is $O(K \cdot G)$ but with precomputed prior.

3.4 Mathematical Justification

Theorem 2 (Consistency of Global Dispersion)**.** *Under the model $Y_{cg} \sim NB(\mu_{cg}, \alpha_g)$ where α_g is perturbation-invariant:*

$$\hat{\alpha}_g^{global} \xrightarrow{p} \alpha_g \quad \text{as } N \to \infty \tag{10}$$

with asymptotic variance:

$$Var(\hat{\alpha}_g^{global}) = \frac{1}{\sum_{c=1}^{N} \mathcal{I}_{cc}(\alpha_g)} \tag{11}$$

where $\mathcal{I}_{cc}$ is the Fisher information from cell c.

Proof sketch. The MLE is consistent for well-specified models. Since the true α_g is shared across perturbations, pooling observations increases Fisher information while maintaining consistency. The asymptotic variance follows from the Cramér-Rao bound. □

4 Size Factor Estimation

4.1 Median-of-Ratios Method

DESeq2 computes size factors as:

$$s_c = \text{median}_g \frac{Y_{cg}}{\left(\prod_{c'=1}^{N} Y_{c'g}\right)^{1/N}} \tag{12}$$

This requires computing the geometric mean across all cells, which is $O(N \cdot G)$.

4.2 Global vs. Per-Comparison Size Factors

Definition 1 (Size Factor Scope)**.**

- **Global**: Compute s_c once from all N cells. Complexity: $O(N \cdot G)$ total.
- **Per-comparison**: Compute $s_c^{(k)}$ for each comparison subset. Complexity: $O(K \cdot N_{\text{subset}} \cdot G)$ total.

For typical CRISPR screens where control cells are shared across all comparisons, global size factors provide:

1. **Consistency**: Same normalization applied to control cells across all comparisons
2. **Stability**: More robust estimates from larger sample size
3. **Speed**: Single computation vs. K computations

4.3 Sparse-Optimized Computation

For single-cell data with $> 90\%$ zeros, we use:

$$s_c = \text{median}_g \frac{Y_{cg}}{r_g} \quad \text{where } r_g = \exp\left(\frac{1}{|\mathcal{C}_g|} \sum_{c:Y_{cg}>0} \log Y_{cg}\right) \tag{13}$$

This avoids computing on zero entries, reducing effective complexity.

5 Implementation in CRISPyx

5.1 Control Statistics Cache

5.2 API Design

```
# CRISPyx API
result = crispyx.de.nb_glm_test(
    "data.h5ad",
    perturbation_column="perturbation",
    dispersion_scope="global",     # NEW: "global" | "per_comparison"
    size_factor_scope="global",    # "global" | "per_comparison"
)
```

Algorithm 1 Global Dispersion Precomputation (Fast Mode)

procedure PRECOMPUTEGLOBALDISPERSION(X, s)
  $\bar{\mu}_g \leftarrow \frac{1}{N}\sum_c Y_{cg}/s_c$ ▷ Normalized mean
  $\hat{\alpha}_g^{\text{MoM}} \leftarrow \frac{\text{Var}(Y_{\cdot g}/s)-\bar{\mu}_g}{\bar{\mu}_g^2}$ ▷ Moments estimate
  Fit trend: $\alpha(\bar{\mu}) = a_0 + a_1/\bar{\mu}$ using gamma GLM
  $\sigma^2_{\text{prior}} \leftarrow \text{Var}(\log\hat{\alpha} - \log\alpha(\bar{\mu})) - \psi'(\text{df}/2)$
  Fast mode: $\hat{\alpha}_g^{\text{global}} \leftarrow \exp\left(\frac{\log\hat{\alpha}_g^{\text{MoM}}+\log\alpha(\bar{\mu}_g)}{2}\right)$ ▷ Simple average in log-space
  return $(\hat{\alpha}^{\text{global}}, \alpha(\cdot), \sigma^2_{\text{prior}})$
end procedure

The `dispersion_scope` parameter controls:

- `"global"`: Precompute dispersion once using all cells (default)
- `"per_comparison"`: DESeq2-style per-comparison estimation (for validation)

5.3 Fast Mode Optimization

The key insight for practical speedup is that full MAP grid search is expensive ($O(G \cdot N \cdot n_{\text{grid}} \cdot n_{\text{refine}})$). We introduce a **fast mode** that replaces MAP with simple trend shrinkage:

$$\log\hat{\alpha}_g^{\text{fast}} = \frac{\log\hat{\alpha}_g^{\text{MoM}} + \log\alpha(\bar{\mu}_g)}{2} \tag{14}$$

This is a geometric mean between the moments-based estimate and the trend, computed in $O(G)$ time. The rationale:

- For genes near the trend: $\hat{\alpha}_g^{\text{MoM}} \approx \alpha(\bar{\mu}_g)$, so the average is close to both
- For outlier genes: shrinkage toward trend prevents extreme values
- Avoids expensive per-gene optimization while maintaining biological plausibility

Complexity comparison:

- Full MAP: $O(G \cdot N \cdot 25 \cdot 20) \approx O(500 \cdot G \cdot N)$ per global estimation
- Fast mode: $O(G \cdot N)$ for MoM + $O(G)$ for averaging $\approx O(G \cdot N)$

This provides $\sim$50$\times$ reduction in global precomputation time while maintaining perfect correlation with per-comparison results.

6 Empirical Validation

6.1 Test Datasets

6.2 Correlation Analysis

We compare global vs. per-comparison estimation:

Key finding: Perfect correlation validates that global dispersion produces *identical* statistical results.

Dataset	Cells	Genes	Perturbations	Source
Adamson_subset	1,716	11,630	3	Adamson et al. 2016
Tian-CRISPRa	20,720	5,943	96	Tian et al. 2019

Table 1: Benchmark datasets for validation

Dataset	LFC ρ	$\log(\alpha)$ ρ	$-\log_{10}(p)$ ρ
Adamson_subset	1.000000	1.000000	1.000000
Tian-CRISPRa	1.000000	1.000000	1.000000

Table 2: Pearson correlation between global and per-comparison results

6.3 Runtime Performance

6.4 Theoretical Speedup Analysis

For a dataset with K perturbations, the theoretical speedup from global dispersion is:

$$\text{Speedup} = \frac{T_{\text{per-comp}}}{T_{\text{global}}} = \frac{K \cdot T_{\text{disp}}}{T_{\text{disp}} + K \cdot T_{\text{GLM}}} = \frac{K}{1 + K \cdot r} \tag{15}$$

where $r = T_{\text{GLM}}/T_{\text{disp}}$ is the ratio of GLM fitting time to dispersion estimation time.

For typical single-cell data where dispersion estimation dominates ($r \ll 1$):

$$\text{Speedup} \approx K \quad (\text{up to } K\times \text{ for large screens}) \tag{16}$$

For $K = 1{,}000$ perturbations, this represents up to **1,000× speedup** in the dispersion estimation phase.

7 Gene-Specific Prior Scaling for LFC Shrinkage

7.1 Motivation

The apeGLM shrinkage applies a Cauchy prior on log-fold changes:

$$p(\beta \mid s) = \frac{1}{\pi s\,(1 + (\beta/s)^2)} \tag{17}$$

Standard apeGLM uses a *global* prior scale s estimated from all genes. However, lowly-expressed genes have higher variance in their LFC estimates, suggesting they should be shrunk more aggressively.

7.2 Gene-Specific Prior Scale

We introduce expression-dependent prior scaling:

$$s_g = s_0 \cdot \left(\frac{\bar{\mu}_g}{\mu_{\text{ref}}}\right)^{-1/2} \tag{18}$$

where $\bar{\mu}_g$ is the mean normalized expression for gene g and $\mu_{\text{ref}} = 100$ is a reference expression level.

Effect:

Dataset	Global (s)	Per-Comparison (s)	Speedup
Adamson_subset	12.93	29.61	2.29×
Tian-CRISPRa	106.35	178.31	1.68×

Table 3: Runtime comparison with `fast_mode=True` global dispersion estimation. Global mode uses MoM + trend shrinkage instead of expensive MAP grid search, enabling substantial speedup while maintaining perfect correlation with per-comparison results.

- Genes with $\bar{\mu}_g = 1$: Prior scale $= s_0 \cdot 10$ (shrunk 10× more)
- Genes with $\bar{\mu}_g = 100$: Prior scale $= s_0$ (standard shrinkage)
- Genes with $\bar{\mu}_g = 10{,}000$: Prior scale $= s_0 \cdot 0.1$ (less shrinkage)

7.3 Hybrid Fallback for Problematic Genes

Some genes have statistics that violate Gaussian likelihood assumptions:

- Large $|\text{MLE}|/\text{SE} > 3$: Skewed posterior
- Low expression $\bar{\mu}_g < 10$: Discretization effects

For these genes (∼5% of total), we flag them for optional full NB-GLM re-fitting:

$$\mathcal{G}_{\text{refit}} = \{g : |\hat{\beta}_g|/\text{SE}_g > 3 \text{ or } \bar{\mu}_g < 10\} \tag{19}$$

8 Conclusion

We have presented a principled approach to global dispersion estimation for single-cell CRISPR screen analysis:

1. **Biological justification**: Dispersion reflects perturbation-invariant technical/biological noise
2. **Mathematical formulation**: Consistent estimation with improved precision from pooled samples
3. **Empirical validation**: Perfect correlation ($\rho = 1.0$) with per-comparison approach
4. **Computational benefit**: $O(1)$ vs. $O(K)$ dispersion fits per screen

The implementation in CRISPyx provides a `dispersion_scope` parameter allowing users to choose between accuracy-focused (per-comparison) and speed-focused (global) modes.

8.1 Future Work

- ~~Full worker-level integration of precomputed global dispersion cache~~ — **Completed**: Workers now skip MoM dispersion and trend fitting when global cache is available
- ~~Fast mode for global precomputation~~ — **Completed**: `fast_mode=True` uses simple trend shrinkage instead of expensive MAP grid search
- Adaptive hybrid mode: use global dispersion for most genes, per-comparison for outliers
- Extension to multi-batch experimental designs

Acknowledgments

This work builds upon the statistical foundations established by DESeq2 (Love et al., 2014) and PyDESeq2 (Muzellec et al., 2023).

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